Today's agenda

- No. of Connected Component
- Topological Sort

AlgoPrep

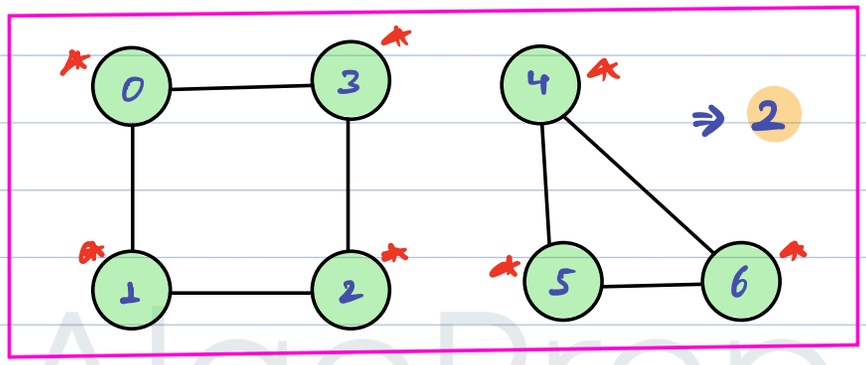
Q) Connected Components

↳ Given undirected graph, find no. of Connected Components.

Note: A Component is said to be Connected, if from every node we can visit all nodes inside that Component.

Ex:

0	1	2	3	4	5	6
T	T	T	T	T	T	T



⇒ 2

graph

0	1	2	3	4	5	6
1	2	1	2	6	4	4
3	0	3	0	5	6	5

AlgoPrep

//Pseudo code

(multi function dfs

```
int main() {
    int n=          int m=
    List<List<Integer>> graph = Construction();

    int ans=0;
    boolean[] vis = new boolean[n];
    for (int i=0; i<n; i++) {
        if (vis[i]==false) {
            vis[i]=true;
            dfs(graph, vis, i);
            ans++;
        }
    }
    return ans;
}
```

T.C: O(V+E)

S.C: O(V)

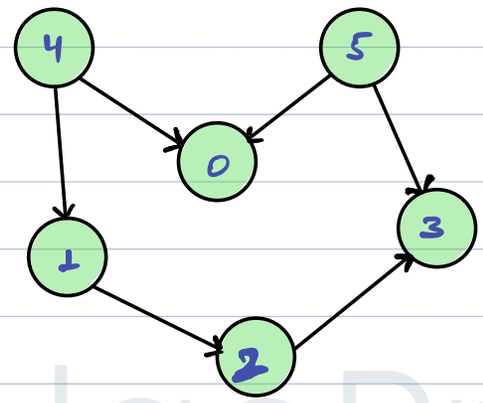
```
void dfs (List<List<Integer>>graph, boolean vis[n], int src) {

    List<Integer> nbrs= graph.get(src);

    for (int v: nbrs) {
        if (vis[v]==false) {
            vis[v]=true;
            dfs (graph, vis, v);
        }
    }

}
```

AlgoPrep

Q) Topological Sort

↳ Give the linear ordering of graph such that for every directed edge (u,v), vertex u comes before v in the ordering. { DAG → Directed, Acyclic graph only }

Ex:



TS: 1 3 2 0 4 5 ✗✗

TS: 4 0 5 1 3 2 ✗✗

These can be multiple topological sort of same graph.

TS: 4 5 0 1 2 3 ✓✓

TS: 5 4 0 1 2 3 ✓✓

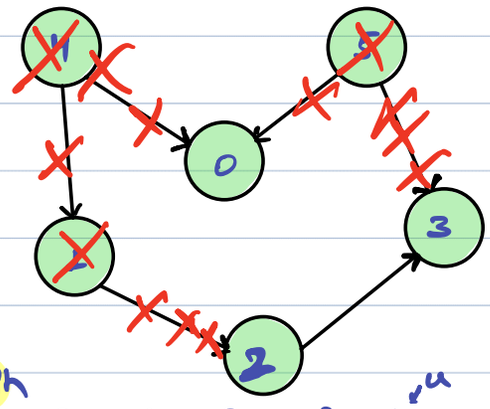
AlgoPrep

// Kahn's algo

no. of incoming edge of a node.

indegree:

0	1	2	3	4	5
~~2~~ 0	~~1~~ 0	~~0~~	~~0~~	0	0



q: ~~4~~ ~~5~~ ~~1~~ ~~0~~ ~~2~~ ~~3~~

graph

0	1	2	3	4 (u)	5
	2	3		1	3
				0	0

Topological sort:

4 5 1 0 2 3

// Pseudo code

```
void topologicalSort (List<List<Integer>> graph,
                      int n, int m) {

    int[] indegree = new int[n];

    for (int u=0; u<n; u++) {
        List<Integer> nbrs = graph.get(u);
        for (int v: nbrs) {
            indegree[v]++;
        }
    }
```

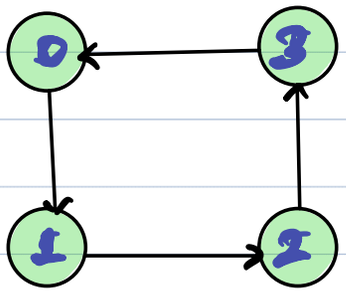
T.C: O(V+E)

S.C: O(V)

```
Queue<Integer> q;

for(int i=0; i<n; i++){
    if(indegree[i]==0){ q.add(i);
}

while (q.size() >0){
    int rem = q.remove();
    s.o.p(rem);

    List<Integer> nbrs = graph.get(rem);

    for (int v: nbrs) {
        indegree[v]--;
        if(indegree[v]==0){
            q.add(v);
        }
    }
}
}
```

AlgoPrep

① what happens if graph is cyclic?

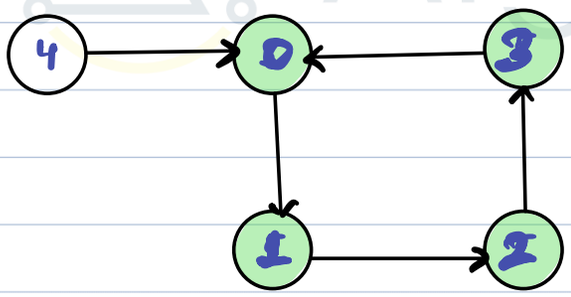


indegree :

0	1	2	3
1	1	1	1

$u \longrightarrow v$

$u \prec\prec v$



indegree :

0	1	2	3	4
~~1~~ 2	1	1	1	0

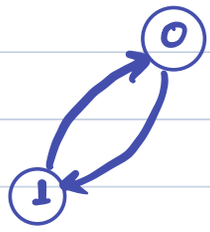
q: ~~4~~

sem=4

4

→ in cyclic graph, dependency is also cyclic. nowhere to start from.

(ii) why not undirected graph?



→ Same issue as cyclic graph.

Q) Detect cycle in directed graph.

directed graph

acyclic
topological sort is valid.

cyclic
↳ you can't complete the topological ordering.

//Pseudo code

```
void topologicalSort (List<List<Integer>> graph,
                      int n, int m) {

    int[] indegree = new int[N];

    for (int u=0; u<N; u++) {
        List<Integer> nbors = graph.get(u);
        for (int v: nbors) {
            indegree[v]++;
```

```
        }
    }
```

```
Queue< Integer> q;
```

T.C: O(V+E)

S.C: O(V)

```
for(int i=0; i<n; i++){
    if (indegree[i]==0){ q.add(i);
}

int count=0;
while (q.size() >0){
    int rem = q.remove();
    ~~s.o.p(rem);~~
    count ++;

    List< Integer> nbrs = graph.get(rem);

    for (int v: nbrs) {
        indegree[v]--;
        if(indegree[v]==0){
            q.add(v);
        }

    }
}
if (count==n){ s.o.p ("acyclic"); }
    else { s.o.p ("cyclic"); }
```

AlgoPrep

3

AlgoPrep